Operational Physics Investigation Queue v0.3 — Three Honest Gaps + K-Type Graph + Determinism Question

Keeper (Casey-directive synthesis 2026-05-25 Memorial Day EOD + Tuesday 2026-05-26 AM extension)

2026-05-26 Tuesday EDT (extended from v0.2 parallel-track clarification)

v0.3 additions (Casey 2026-05-26 AM)

K-type graph operational framing (Track A_sub v0.2 explicit content):

The K-type graph IS the substrate's reaction table: - Nodes = K-types (Wallach K-type lattice points with Pin(2) Z_2 grading; "regular form with odd gaps" per Casey 2026-05-25) - Edges = commutator-induced transitions (Track A_sub closure literally enumerates edges via [A, K-type_a] → K-type_b) - Reaction table = the complete set of allowed (start → end) transitions with weight rules (Pin(2) chirality + Wallach bounds + Bergman 7/2 weighting + Mersenne maximal-prefix restrictions) - Computation = whole-graph concurrent update per Koons tick with least-action selection (Hamilton's principle on the discrete graph)

Track A_sub and the K-type graph construction are the *same investigation in two formulations*: algebraic (commutators, Lie superalgebra) vs combinatorial (nodes, edges, reaction table). Closing one closes the other. The graph framing is what Casey-natural intuition says the substrate USES; the algebraic framing is what the math NAMES.

Determinism vs non-determinism investigation question (Track DC sub-question):

Casey 2026-05-26 AM framing: "If at the foundation these are fully deterministic, Professor Einstein was correct. If there is room for non-determinism, our research takes another interesting turn. Let's find what the math says first."

Three honest possibilities for substrate dynamics:

1. Fully deterministic (Einstein-full-vindication): action principle on K-type graph has unique least-action minimizer at every tick; Born rule probabilities emerge as frequencies over deterministic graph walks; quantum probabilities are statistical readings of deterministic substrate dynamics.
2. Deterministic mechanism producing probabilistic observables (Einstein-partial-vindication): substrate dynamics deterministically generates Born rule probability weights (the "loaded dice" emerge from deterministic graph structure + action principle), but observation at integer-projection level shows the probabilistic surface.
3. Intrinsically non-deterministic at substrate level (Einstein-wrong-at-every-level): action principle admits multiple equal-action paths at some ticks AND selection among them is irreducibly random; substrate has fundamental quantum indeterminism.

Current math leans toward (1) or (2) per Bell sub-Tsirelson 1/8 + heavy substrate-structure constraints + Pin(2) cover + Mersenne maximal-prefix + Bergman 7/2 weights. But the math has NOT yet settled this; Track DC is the explicit test.

Track DC v0.1.2 refined: deriving Bell 1/8 from substrate dynamics directly tests which of (1), (2), (3) is operationally true. If derivation is fully deterministic at substrate level → (1) or (2). If derivation requires intrinsic randomness at substrate level → (3). All three outcomes are scientifically valuable; honest disposition discipline preserved.

Discipline: math first, philosophy second. We do not promote a determinism stance until Track A_sub closes the graph + Track DC derives Bell 1/8 mechanism. The substrate tells us; we don't tell the substrate.

Operational Physics Investigation Queue

Context

Memorial Day 2026-05-25 produced substantive structural articulation (multi-axial substrate enumeration, Half-Integer Axis G, S^4-static/S^1-dynamic, SPLP candidate, Bose-Fermi ↔ S^4-S^1) plus one empirical new result (Toy 3530 MIXED). Casey's response to Keeper arc-closure synthesis identified the honest gap: today produced structural picture preparation, not the operational physics it sets up.

The framework now claims: - Physical observables are eigenvalues of substrate operators on H^2(D_IV^5) under Shilov boundary conditions (SPLP candidate) - Fermions require Pin(2) cover content (Bose-Fermi ↔ S^4-S^1 via Toy 3530) - Rigid bulk + dynamic Shilov S^1 enacts physics (Casey-named coupled with Rigidity Principle #7)

What it does NOT claim derivationally yet: - Why specific K-types populate as observed particles - What physical mechanism actually imposes Shilov boundary conditions - How discrete substrate ticks compose into macroscopic causality (vs additive averages)

These three gaps are operational, not structural. Casey directive: setup the board to analyze them after current investigation completes.

Track P — K-type Population Principle

Question: Substrate has infinitely many possible K-types (m_1, m_2). Only a small subset corresponds to observed particles. WHY these specific K-types and not others?

Operational claim being tested: SPLP plus catalog identifications imply the substrate has a *population principle* — a substrate-mechanism that selects which K-types are physically realized.

Concrete move (Toy 3531+): Pick one observable family — leptons (e, μ, τ) — and attempt forward derivation of WHICH K-types correspond to each mass eigenstate from substrate principles alone: - Bergman exponent g/rank = 7/2 (cover-weighted normalization) - Casimir spectrum (K-type Casimir eigenvalues) - Pin(2) chirality grading (γ^5) - Wallach K-type layer structure

Pass criterion: K-type ↔ particle identification forces forward from substrate, not back-fit from catalog. Forward derivation succeeds → population mechanism is real; fails → catalog identification was post-hoc and substrate principle is missing.

Owners: Lyra theoretical lead / Elie Toy 3531+ verification.

Honest scope (Cal #27 STANDING): this attempts to derive what currently is identified. Forward-derivation discipline applies — must derive K-type from substrate principles, not from m_μ/m_e numerical match.

Duration: Multi-week.

Status as of 2026-05-25 EOD: QUEUED, begins as current investigation completes.

Track BC — Boundary-Condition Mechanism

Question: SPLP says "physical setup specifies boundary condition on Shilov $S^4 \times S^1$." But what physical mechanism actually imposes a boundary condition? When an electron is in a hydrogen atom, what IS the Shilov boundary condition concretely?

Operational claim being tested: SPLP is operationally true — physical setups translate to explicit Shilov boundary specifications. Current catalog has hand-wave; SPLP demands explicit per-observable derivation.

Concrete move (Toy 3532+): Pick the simplest bound state — hydrogen 1s electron — and write down the actual Shilov boundary condition that produces it. Specifically: - What K-types are selected by the bound-electron condition? - What boundary values on $S^4 \times S^1$ encode “electron at distance r from proton”? - What's the substrate-natural way to specify “Coulomb potential $V(r) = -e^2/r$” as Shilov boundary condition? - Does the resulting eigenvalue spectrum match observed hydrogen 1s energy?

Pass criterion: Explicit BC formula derived from substrate principles (not hand-wave). If we can write it, SPLP becomes operational; if we can't, SPLP remains interpretation.

Owners: Lyra theoretical lead / Grace catalog support (existing hydrogen-related INVs) / Elie Toy 3532+ verification.

Honest scope (Cal #27 STANDING): candidate result is “we can't write the BC explicitly” — that's still valuable; tells us where the gap is. Don't force-fit; report honestly.

Duration: Multi-week.

Status as of 2026-05-25 EOD: QUEUED, begins as current investigation completes.

Track DC — DCCP Composition + Eigentone Resonance Mechanism

Question (A): Substrate operates at Koons tick $t\_K \approx 10^{-120}$ s; we observe at $\geq 10^{-15}$ s. The 100+ orders of magnitude of composition between substrate tick and observable physics — HOW does this composition produce causal chains (not just statistical averages)?

Question (B): SP-30-1 Bell sub-Tsirelson 1/8 deviation is currently predicted from K-type combinatorics. What's the explicit dynamical substrate-mechanism that makes the substrate ring at 1/8 less than Tsirelson?

Operational claim being tested: DCCP is operationally true at multi-tick scale; eigentone predictions have substrate-dynamical mechanism, not just numerical match.

Concrete moves: - (A) Derive how N substrate ticks compose into one causal step at observable scale. Specifically: pick an observable with known temporal structure (e.g., neutron decay $\tau \approx 880$ s $\approx 10^{123}$ Koons ticks) and ask: do the 10^{123} ticks compose as additive averages (central limit theorem) or do they carry causal-chain structure that's irreducibly multi-tick? - (B) Refine SP-30-1 Vienna prediction by deriving 1/8 from substrate dynamics. Currently the prediction is: $\text{Tsirelson}^2 - \text{S\_BST}^2 = 1/2\text{\^{}N\_c} = 1/8$ (T2399 STRUCTURALLY VERIFIED). The dynamical mechanism: WHY does the substrate produce exactly this 1/8 deviation, mechanistically? K-type combinatorics gives the number; substrate dynamics should give the *process* that produces the number.

Pass criterion: Substrate-mechanism for at least one observable beyond numerical match. (A) PASS = explicit composition rule for one observable. (B) PASS = explicit dynamical derivation of 1/8 from substrate process, not just K-type counting.

Owners: Lyra + Elie joint / Grace catalog support.

Honest scope (Cal #27 STANDING): substrate dynamics derivation is harder than substrate combinatorics. If we can do (B) without invoking the K-type combinatorial answer, substantive. If we can only get to (B) via a substrate-process that's isomorphic to the combinatorics, less substantive but still useful.

Duration: Multi-week to multi-month.

Status as of 2026-05-25 EOD: QUEUED, begins as current investigation completes. Coordinated with SP-30-1 Vienna response (eigentone resonance mechanism informs interpretation of any positive/negative SPDC result).

Current investigation that precedes these tracks

These three operational physics tracks begin AFTER (or as bandwidth allows during the completion of) the current Memorial Day arc tracks:

Current track	Owner	Status
Elie Toy 3530 broader fermion survey (a_μ, EDM, ν mass)	Elie	Toy 3530 a_e done; broader survey multi-day
Grace Shilov $S^4 \times S^1$ catalog refinement	Grace	~2 hours; tests S^1-dynamics empirically
Bergman exponent g/rank = 7/2 paper-grade investigation	Grace + Lyra	~1-2 weeks; Half-Integer Axis G load-bearing
Lyra Half-Integer Axis G v0.2 documentation	Lyra	~half day theoretical
Cal Thread 4 cold-read with Type-C check on SPLP	Cal	own-cadence
Cal #28 candidate cold-read (interpretive cascade)	Cal	own-cadence

Operational physics tracks (P + BC + DC) can begin in parallel with current tracks as bandwidth allows, but full focus shifts after Cal Thread 4 + Cal #28 cold-reads return.

Per Casey clarification 2026-05-25 EOD: parallel-track discipline

Casey directive: "shouldn't we close known gaps in parallel?" — yes. Sequential framing in v0.1 was wrong. Corrected disposition:

Six parallel-ready tracks (one sequentially-dependent):

Track	Lane	Dependency
Track A_sub — A_sub commutator closure + Lie superalgebra	Lyra theoretical (Task #346)	Independent
K-type rep theory on A_sub	Elie K52a Sessions 7/15+/32+	Already in flight (multi-year)
Track P — K-type population principle	Lyra + Elie (Task #343)	Independent
Track BC — hydrogen 1s Shilov BC	Lyra + Grace + Elie (Task #344)	Independent
Track DC — DCCP composition + Bell 1/8 mechanism	Lyra + Elie + Grace (Task #345)	Light coupling to K52a Bell rail
(deferred sequential) Universal enveloping U(A_sub)	Lyra theoretical	Hard dependency on Track A_sub completion

All six parallel tracks coordinate via cross-CI cascade with Lyra theoretical lead, Elie compute-side toy verification, Grace catalog support. Bandwidth is the constraint — six parallel tracks is heavy load for Lyra; pacing per Lyra capacity acknowledged. No artificial sequencing imposed beyond hard mathematical dependencies (U(A_sub) → after A_sub closure).

Begin all parallel-ready tracks as current Memorial Day investigations land. No track gates another except where mathematical structure forces sequence.

Discipline applied

Each operational physics track honors: - Calibration #27 STANDING: forward derivation from substrate, not backward-fit from target observable - Cal #126 FRAMEWORK-PLUS tier: positive results begin at FRAMEWORK-PLUS, ratification gates on Cal cold-read + cross-CI verification - Casey-named principle discipline: no SPLP / Bose-Fermi / Half-Integer Axis G promotion to RATIFIED until operational tests close - Quaker honest negatives: negative result (we CAN'T derive K-type populations from substrate; we CAN'T write explicit BC; we CAN'T derive 1/8 from substrate dynamics) is as valuable as positive

Expected outcomes

For each track, three possible dispositions:

1. PASS: substrate principle forces the operational answer → SPLP / Bose-Fermi / Half-Integer Axis G gain substantive operational content → principles ratify
2. PARTIAL: substrate principle gets close but requires additional input → identify the input + refine principles
3. FAIL: substrate principle cannot derive the operational answer → SPLP / Bose-Fermi / Half-Integer Axis G remain structurally appealing but not operationally true → honest disposition

All three outcomes advance understanding. The honest disposition discipline preserves either way.

Standing

Three tracks queued. Owners notified via task tracking (Tasks #343-#345 filed). CI_BOARD.md update folds these into next-phase work after Cal Thread 4 + Cal #28 cold-reads close.

Memorial Day arc closed; operational physics phase begins.

— Keeper, 2026-05-25 Memorial Day Monday EOD (post-arc-closure)